

NITHISH MARNENI

Embedded Software Developer

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PROFESSIONAL SUMMARY

Embedded Software Developer with **4+** years of experience delivering validated, production-grade software in semiconductor and systems engineering environments. Consistently reduces defect escape rates, accelerates validation cycles, and improves system reliability through **Python** automation and low-level **Linux** debugging. Holds an M.S. in Computer Science (4.0 GPA) and a B.Tech in Electronics & Communication, bridging **firmware**-level hardware understanding with full-stack system software skills.

CORE TECHNICAL SKILLS

Languages: C, C++, Embedded C, Python, Bash, SQL

Embedded & OS: Microcontrollers, RTOS Concepts, Device Drivers, Embedded Linux, Board Bring-Up, Memory Management, Multithreading

Protocols: UART, SPI, I2C, CAN, TCP/IP, REST APIs

Debug & Test: GDB, Linux Debugging, Unit Testing, Log Analysis, Root Cause Analysis, Performance Profiling

Signal Processing: DSP, FFT, Filters, Sensor Signal Processing, MOSFET & Amplifier Analysis

Tools & Cloud: Git, Docker, CI/CD, AWS, Jupyter, NumPy, SciPy, Pandas, Matplotlib

PROFESSIONAL EXPERIENCE

Embedded Software Developer · **AMD**

Los Angeles, CA | June 2024 – Present

- Automated validation test harnesses in **Python**, cutting manual **regression** time by **~40%** and enabling the team to validate **3×** more **firmware** builds per sprint.
- Identified and **root-caused 12+** high-severity **Linux**-layer defects by correlating multi-source log traces, reducing mean-time-to-resolution from days to hours.
- Built a data-collection and reporting pipeline that aggregated performance counters across **50+** test nodes, surfacing system bottlenecks **60%** faster than manual analysis.
- Hardened **hardware-software integration** workflows by reproducing intermittent system faults under controlled conditions, closing 8 previously unresolved tickets within the first **quarter**.
- Contributed to 3 consecutive release-readiness milestones on schedule by driving systematic defect triage, **log analysis**, and cross-team coordination.
- Authored internal documentation and code-review guidelines adopted by **5+** engineers, reducing onboarding time for new team members by an estimated **30%**.

Software Engineer · **Inventive Core Integrations**

Hyderabad, India | July 2021 – August 2023

- Delivered 6 automation and backend modules in **Python** that collectively eliminated **~15** hours/week of manual engineering effort across validation and monitoring workflows.
- Debugged and resolved **20+** system-level defects in **Linux** environments, improving deployment reliability by reducing post-release incidents by **35%**.
- Designed a reusable scripting framework for system monitoring and validation that was adopted as a standard tool across 3 concurrent projects.
- Integrated **REST APIs** into data-processing pipelines, enabling real-time ingestion of external system events and reducing data latency from batch (hourly) to near-real-time.
- Improved code quality metrics by **25%** (measured by static analysis tooling) after introducing structured code-review practices and internal technical documentation.

KEY PROJECTS

ElecPy — Electronics & Signal Processing CLI Tool — *Python · NumPy · Matplotlib · FFT · DSP*

- Built a CLI simulation tool covering Ohm's Law, MOSFET drain current, amplifier gain, and **FFT** visualization; reduced circuit calculation iteration time by **70%** vs manual derivation.
- Implemented sampling and frequency-domain analysis modules enabling sub-millisecond **signal** simulation, directly applicable to embedded sensor **firmware** validation workflows.

EMG Pattern Recognition & Signal Processing Pipeline — *Python · SciPy · Pandas · Digital Filters · Feature Extraction*

- Processed raw biomedical EMG **signals** with **notch/band-pass** filters achieving **>95%** noise reduction, directly mirroring embedded sensor **signal** conditioning challenges.
- Extracted **18** time-domain and frequency-domain features via a sliding-window engine, producing a structured dataset that fed downstream classification models with **91%+** accuracy.

Distributed File System — *Python · Linux · REST APIs · Replication · Caching · MongoDB*

- Architected a 6-component distributed file system (registry, directory, file, lock, cache servers) handling concurrent client access with zero data corruption across **200+** test scenarios.
- Implemented replication and locking protocols that maintained consistency under simulated node failure, demonstrating fault-tolerance patterns applicable to embedded distributed systems.

EDUCATION

M.S., Computer Science — Lamar University, Beaumont, TX *July 2023 – May 2025 | GPA: 4.0 / 4.0*

Outstanding Graduate Student Award · Coursework: Distributed Systems, OS, Embedded Systems, DSP, Computer Architecture

B.Tech., Electronics & Communication Engineering — JNTUH, Hyderabad, India *July 2018 – April 2022*